INCH-POUND

A-A-50734B
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SUPERSEDING
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COMMERCIAL ITEM DESCRIPTION

MULTIMETER, RUGGEDIZED

The General Services Administration has authorized the use of this commercial item description by all federal agencies.

1. SCOPE. This commercial item description (CID), describes a Ruggedized Multimeter. This CID is meant as a minimum requirement for the Ruggedized Multimeter in which only those manufacturers that meet or surpass the following requirements are supplied per this CID.

2. SALIENT CHARACTERISTICS. The equipment covered by this CID shall be commercially available and may be modified to the extent necessary to meet this description. The equipment shall be Class 2 in accordance with MIL-PRF-28800, except as specified herein. The equipment shall be bid sample tested to ensure that operation is within the accuracy, limits, and specifications herein.

2.1 Electrical power sources. The only power from which the multimeter shall operate is the power supplied by the battery contained within the multimeter.

2.1.1 Battery. One 9V battery, National Electronic Distributors Association (NEDA) type 1604, 9V alkaline, or equal, shall be used within the multimeter. In voltage or current measurement functions (under worse case battery drain conditions), the battery shall be capable of sustaining 600 hours of operation at 25°C and 300 hours of operation at –20°C or 55°C. In resistance or diode test functions (under worst case battery conditions), the battery shall be capable of sustaining 300 hours of operation at 25°C and 250 hours of operation at -20°C or 55°C. The battery may be shipped installed.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data, which may improve this document, should be sent to: WR-ALC/LEEC, 295 Byron St., Robins AFB, GA 31098-1611.

AMSC N/A FSC 6625

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2.1.2 Battery test. A display indication shall be provided to show marginal performance of the multimeter due to low battery potential.

2.2 Temperature requirements.

2.2.1 Non-operating (storage) temperature. The equipment shall be allowed to be stored in a non-operating environment of -51°C to 71°C without damage to the equipment.

2.2.2 Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of -15°C to +55°C, -40°C for 20 minutes when taken from 20°C.

2.3 Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be at least 2,000 hours.

2.4 Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effect of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

2.5 Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration and maintenance adjustments. These adjustments shall be provided by variable value components, which are adjustable, by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules. The calibration interval shall be a period of one year or greater based on an operating time of 2,000 hours.

2.6 Performance characteristics.

2.6.1 Accuracy. Unless otherwise specified, all electrical specifications in paragraph 2.6 are from 18°C to 28°C with up to 95 percent humidity, for one year after calibration. At temperature less than 18°C or greater than 28°C, the temperature coefficient shall be less than 0.1 times the applicable electrical specification per °C. Accuracy is specified as ± ([percent of reading] + [number of least significant digits]).

2.6.2 DC voltage.

2.6.2.1 Ranges, resolution, and accuracy. Minimum performance shall be as specified in Table I.

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TABLE I. DC Voltage.

RANGE	RESOLUTION	ACCURACY
320.0 mV	100 µV	± (0.2% +1 LSD*)
3.200 V	1 mV	± (0.2% +1 LSD)
32.00 V	10 mV	± (0.2% +1 LSD)
320.0 V	100 mV	± (0.2% +1 LSD)
1000 V	1V	± (0.2% +1 LSD)

* Least Significant Digit.

2.6.2.2 Input Impedance. The input impedance shall be greater than 9.9 Megohms.

2.6.2.3 Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 1,000 Vrms on all ranges, except 500 Vrms continuously on the 300.0 mV range.

2.6.2.4 Normal mode rejection ratio. Normal mode noise rejection shall not be less than 60 decibels (dB) at 50 and 60 Hz (±0.15%).

2.6.2.5 Common mode rejection ratio. Common mode rejection, with a 1,000 Ohms unbalanced in the low lead, shall not be less than 120 decibels (dB) at 50 and 60 Hz (±0.15%).

2.6.3 AC Voltage.

2.6.3.1 Range, resolution, and accuracy. Minimum performance shall be as specified in Table II.

TABLE II - AC Voltage.

RANGE	RESOLUTION	ACCURACY		
		40 Hz to 2 KHz	2 KHz to 10 KHz	10 KHz to 30 KHz
320.0 mV	100 µV	±(0.5%+3 LSD)	±(2%+3 LSD)	±(4%+10 LSD)
3.200 V	1 mV	±(0.5%+3 LSD)	±(2%+3 LSD)	±(4%+10 LSD)
32.00 V	10 mV	±(0.5%+3 LSD)	±(2%+3 LSD)	±(4%+10 LSD)
320.0 V	100 mV	±(0.5%+3 LSD)	±(2%+3 LSD)	±(4%+10 LSD)
1000 V	1V	±(1% + 3 LSD)	±(3%+3 LSD)	Not to exceed 15%

2.6.3.2 Input impedance. The input impedance shall be greater than 9.9 Megohms.

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2.6.3.3 Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 1,000 Vrms on all ranges, except 500 Vrms continuously on the 300 mV range. The multimeter shall be capable of withstanding an AC Volt-Hertz product up to 10^7.

2.6.3.4 Common mode rejection ratio. Common mode rejection, with a 1,000 ohms unbalance in the low lead, shall not be less than 60 decibels (dB) at 50 and 60 Hz (±0.15%).

2.6.4 Resistance and Conductance.

2.6.4.1 Ranges, resolution and accuracy. Minimum performance shall be as specified in Table III.

TABLE III. Resistance and Conductance.

RANGE	RESOLUTION	ACCURACY
320.0 ohm	0.1 ohm	± (0.3%+3 LSD)
3.200 Kohm	1 ohm	± (0.2%+1 LSD)
32.00 Kohm	10 ohm	± (0.2%+1 LSD)
320.0 Kohm	100 ohm	± (0.2%+1 LSD)
3.200 Mohm	1 Kohm	± (0.2%+1 LSD)
32.00 Mohm	10 Kohm	± (1%+1 LSD)
32.00 nS*	0.01nS	± (2%+10 LSD)

* nano-Siemen (Mohms = 1000/nS)

2.6.4.2 Humidity. The 32.00 nS range accuracy specification shall be limited to 70% maximum relative humidity from 28°C to 55°C.

2.6.4.3 Maximum current through unknown resistance. The multimeter shall not provide more than 1.5 mA on any range.

2.6.4.4 Maximum open terminal voltage. The open terminal voltage shall not be greater than 3.0 volts on any range.

2.6.4.5 Maximum on scale test voltage. The multimeter shall be capable of allowing in-circuit resistance measurements. It shall not apply greater than 450 mV at full scale on the following ranges: 320 ohms, 3.2 Kohms, 32 Kohms, 320 Kohms, and 3.2 Mohms.

2.6.4.6 Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 500 Vrms on all ranges.

2.6.5 Diode and continuity test.

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2.6.5.1 Continuity test. The multimeter shall provide a continuous audible tone (or equivalent) for test resistance below 100 ohms. Forward biased (0.15 volts or greater) diode or transistor junctions shall inhibit continuous tone. Any deviation requires Air Force approval.

2.6.5.2 Diode test. The multimeter shall be capable of displaying diode or transistor junction voltage drops to at least 2.000 volts. A momentary audible tone shall indicate a forward biased junction, and the absence of sound shall indicate a reversed biased or open junction.

2.6.5.3 Maximum test current. The maximum diode or continuity test current shall not exceed 1.5 mA.

2.6.5.4 Maximum open terminal voltage. The open terminal diode or continuity test voltage shall not be greater than 3.5 volts.

2.6.5.5 Overload. The multimeter shall be capable of continuously withstanding, without damage or blown fuses, 500 Vrms on all ranges.

2.6.6 DC and AC current.

2.6.6.1 Ranges, resolution, accuracy, and burden voltage. Minimum performance shall be as specified in Table IV.

TABLE IV. DC and AC Current.

RANGE	RESOLUTION	DC ACCURACY	AC ACCURACY 40 Hz TO 1 KHz	MAX BURDEN (V)
320.0 µA	0.1 µA	±(0.75%+2 LSD)	±(1.5%+2 LSD)	160 mV
3200 µA	1 µA	±(0.75%+2 LSD)	±(1.5%+2 LSD)	1.6 V
32.00 mA	10 µA	±(0.75%+2 LSD)	±(1.5%+2 LSD)	180 mV
320.0 mA	100 µA	±(0.75%+2 LSD)	±(1.5%+2 LSD)	1.8 V
10.00 A	10 mA	±(0.75%+2 LSD)	±(1.5%+2 LSD)	0.5 V

2.6.6.2 Overload. The multimeter shall be capable of continuously withstanding; without damage,
0.4 Vrms on all ranges; except the 10 A range shall be capable of withstanding 10 Vrms continuously and 20 A for 30 seconds. All current inputs shall be fused or protected to interrupt overload currents, up to 10,000 A, from accidental voltage overloads in circuits up to 600 volts.

2.6.7 Controls and visual display.

2.6.7.1 Ranges and function control. All input functions (voltage, current, resistance, and diode/continuity test) shall be controlled with one rotary switch. Ranging shall be automatic (with a manual override) in DC and AC voltage from the 3 V range to the 1,000 V range, and in

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resistance from the 300 ohm range to the 30 Mohms range. Ranging shall be automatic (with a manual override) in DC and AC current between 30 mA and 300 mA; and between 300 µA and 3,000 µA.

2.6.7.2 Readout display. The readout display shall be a minimum of 3 ½ digits with a digital readout with a minimum update rate of 3,000 samples/sec, an analog bar-graph (or equivalent) readout with at least 30 segments of resolution, and auto-zeroing. The digital display shall update at least 2 times per-second and the analog display shall update at least 20 times per-second.

2.6.7.3 Automatic polarity. The polarity of the measured value shall be automatically indicated on the digital and analog displays.

2.6.7.4 Response time. Response time is defined as the time interval between an abrupt change of input and the time the new value is displayed within specified accuracy, including the worst case time to change ranges in the auto range mode.

2.6.7.5 Direct current or voltage response. Direct current or voltage response shall be less than 2 seconds.

2.6.7.6 Alternating current or voltage response. Alternating current or voltage response time shall be less than 4 seconds.

2.6.7.7 Resistance response time. Resistance response time shall be less than 2 seconds on all ranges except that in the nano-Siemen range or the lowest and highest resistance ranges, it shall not be more than 6 seconds.

2.6.7.8 Reading hold. The multimeter shall be capable of holding a reading in the display, for the operator to view, after the test leads have been removed from the device under test.

2.6.8 Accessories.

2.6.8.1 Test leads. Two test leads (one red, one black) shall be supplied with the multimeter. The test leads shall be at least 1 meter long, rated to withstand at least 1,000 volts, and have right angle safety connectors.

2.6.8.2 Meter case. Non-metallic case, ruggedized to withstand harsh environment normally found in flight line applications.

2.6.9 Maintenance and calibration accessories. If required, the multimeter shall be provided with all necessary maintenance and calibration accessories. Not more than one circuit board extender or patch cable shall be required to permit maintenance of the modules and plug-in circuit cards.

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3. PRODUCT CONFORMANCE PROVISIONS.

3.1 Product Conformance. The products provided shall meet the salient characteristics of this Commercial Item Description, conform to the producer's own drawings, specifications, standards and quality assurance practices, and be the same product offered for sale in the commercial market. The government reserves the right to require proof of such conformance.

3.2 Metric Products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within the specified tolerances using conversion tables contained in the latest revision of Federal Standard 376, and all other requirements of this CID are met. If a product manufactured to metric dimensions exceeds the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

3.3 Market Acceptability. The following acceptability criteria are necessary to document the quality of the product to be provided under this CID.

3.3.1 The manufacturer of the item must have been producing a product meeting the requirements of this CID for at least 2 years.

3.3.2 The company must have sold at least 1000 units meeting this CID in the commercial market place over the past 2 years.

4. PACKAGING. Preservation, packing, and marking shall be as specified in the contract or order.

5. NOTES.

5.1 Contractor Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description and that the product conforms to the manufacturer's own drawings, specifications, standards and quality assurance practices. The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

5.2 Operation and Maintenance Manuals. The contractor shall supply, at no cost to the Air Force, two operation and maintenance manuals. These shall be placed within the same carton as the instrument itself.

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5.3 Regulatory Requirements. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR); the Government's policy is to acquire items composed of the highest percentage of recovered materials where practical. These are consistent with maintaining a satisfactory level of competition without adversely affecting performance requirements or exposing the supplier's employees to undue hazards from the recovered materials.

5.4 Sources of documents.

5.4.1 Government documents. Copies of specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Ave, Bldg 4, Section D, Philadelphia, PA 19111-5094.

5.4.2 Code of Federal Regulations (CFR). Copies may be obtained from the Superintendent of Documents, US. Government Printing Office, Washington DC 20402.

5.4.3 NEDA Standards. Copies can be obtained from NEDA, 1111 Alderman Drive, Suite 400, Alpharetta, GA 30005-4143, U.S.A. Phone: 678-393-9990. Website: www.nedassoc.org.

5.4.4 Acceptable manufacturers. The following brand name, model and national stock number are known to be acceptable and current manufacturer is listed below:

Manufacturer	Model	National Stock Number
John Fluke Mfg Co.	Fluke-27	6625-01-262-4815
Procurement Document	A-A-50734B	6625-01-079-1762

5.5 Ordering data. The contract or order should specify the following:

a. CID document number and revision.
b. Product conformance provisions.
c. Packaging requirements.

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MILITARY INTERESTS:

Custodian:

Army - MI
Navy - SH

Reviewer:
Army - CR
DLA - CC
OSD/SO

Preparing Activity:
Air Force - 84

Agent:
Air Force - 99

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